

AHNAF SHAHRIAR

[Email](#) | [LinkedIn](#) | [Github](#)

EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering

Waterloo, ON

Sept. 2021 – Apr. 2026

- **GPA:** 3.7
- **Relevant Courses:** Computer Architecture, Distributed Computing, ML for Chip Design, Robot Dynamics & Control, Compilers, Real-Time Operating Systems

EXPERIENCE

Autopilot AI HW Engineer (Post-Silicon Validation)

Aug. 2025 – Dec. 2025

Tesla

Palo Alto, CA

- **Hardware-to-software bringup:** Led hardware-to-software bringup for Autopilot AI accelerators using FPGA prototypes and virtual platforms to enable early driver and firmware integration before mass silicon roll-out.
- **Emulation & debug:** Built and maintained Python tools workflows (virtual platforms and FPGA in the loop), and developed JTAG and trace capture pipelines to enable root-cause analysis.
- **Post-silicon validation:** Designed and ran automated validation suites and regression harnesses that exercised AI inferencing datapaths and caught corner-case bugs during bringup.
- **Performance & power characterization:** Profiled end-to-end software stacks, identified microarchitectural issues, and implemented workarounds to stabilize performance and power during early bringup.

Wireless Firmware Developer

Jan. 2025 – Apr. 2025

ON Semiconductor Corporation

Waterloo, ON

- **SDK Development:** Integrated *I3C* drivers and *Bluetooth Low Energy* profile services into sample code.
- **Wireless Connectivity:** Developed and integrated cutting-edge *LE Audio*, *HFP*, and *A2DP* functionality into an all-in-one Application.
- **Wireless Manager** Designed a modular state machine for bluetooth, improving power efficiency by *25%*
- **Test Automation:** Automated SDK release test cases, reducing testing time by up to *80%* for samples

Embedded Software Engineering Intern

Sept. 2022 – Dec. 2022

Synapse Product Development

Seattle, WA

- **Prototyping:** Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs:** Designd custom lab automation APIs for equipment from *Agilent*, *Keysight*, *NI*, *Tektronik*.
- **Automation:** Streamlined testing and in house procedures using *Python* and *Bash*.
- **Driver Development:** Wrote low-level *I2C* and *UART* drivers for in-house PCB test platforms on Zephyr RTOS.

Firmware Developer

Jan. 2022 – April 2022

Ford Motor Company of Canada

Remote

- **Unity/Cmock Test framework:** Lead developer for optimization for unit testing, achieving up to *30% faster* runtime while using *50%* less manually written test cases.
- **Automation:** Improved *Jenkins CI/CD* pipelines to support unit testing automation using *Python*
- **Embedded Trace Debugging:** Built high speed Parsing tools to parse through CAN and Serial messages exceeding *10 million+* lines.

PROJECTS

LC-3 Emulator: A C emulator for an educational ISA. Improves by *25%* on research paper using *Python* data logging.

Real Time Executable: A RTOS implementation in STM32 capable of Pre-emptive task switching and its own Malloc

Autonomous Seeker: An Autonomus RC vehicle equipped with a pick and place robot. Created to Compete in Quidditch!

VHDL Compiler: A Java Compiler for creating VHDL circuits. Using a boolean intermediate representation

TECHNICAL SKILLS

Languages: C/C++, Java, Python, Tcl, Bash scripting, ASM, VHDL, SystemVerilog/Verilog

Tools: OpenCV, ROS, Linux, Qemu, LLDB/GDB, AWS, Docker, WireShark, UVM, Matlab

Hardware: Microcontrollers (ESP32, STM32, nRF, Arduino, PIC), Oscilloscopes, Logic Analyzer, Spectrum Analyzer

Protocols: TCP/IP, Ethernet, CAN/CAN-FD, LIN, AXI, Serial(I2C, SPI, I3C, UART), BLE, LoRan